厦门大学《概率统计II》课程试卷

________学院______系______年级______专业

主考教师：______ 试卷类型：（B 卷）

已知标准正态分布的分布函数值如下：$\Phi(1)=0.8413, \Phi(2)=0.9772, \Phi(3)=0.9987$

1. (30分) 设二维随机变量(X,Y)的概率密度函数为

$$f(x,y)=\begin{cases} x+y, & 0<x<1, 0<y<1; \\ 0, & \text{其他}. \end{cases}$$

(1) 求(X,Y)对应的分布函数值$F(0.5,0.5)$以及求概率$P(Y<X)$;

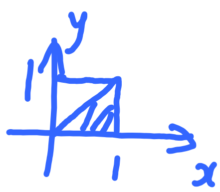
(2) 求边缘概率密度$f_X(x), f_Y(y)$;

(3) 求条件概率密度$f_{Y|X}(y|x)$; 求在$Y=0.5$条件下X的条件概率密度$f_{X|Y}(x|0.5)$.

解：(1) $F(0.5,0.5)=P\{0<X\leq 0.5, \ 0<Y\leq 0.5\}$

$$=\int_0^{0.5}\int_0^{0.5}(x+y)\,dx\,dy$$

$$=0.5\cdot\frac{1}{2}\cdot 0.5^2 \times 2 = \frac{1}{8}=0.125$$



$$P(Y<X)=\int_0^1 dx\int_0^x (x+y)\,dy$$

$$=\int_0^1 x\cdot x+\frac{1}{2}x^2\,dx$$

$$=\frac{3}{2}\cdot\frac{1}{3}=\frac{1}{2}=0.5$$

(2) $f_X(x)=\int_{-\infty}^{+\infty}f(x,y)\,dy$

$$=\begin{cases}\int_0^1(x+y)\,dy=x+\frac{1}{2}, & 0<x<1\\ 0, & \text{其他}\end{cases}$$

$$f_Y(y)=\begin{cases}\int_0^1(x+y)\,dx=y+\frac{1}{2}, & 0<y<1\\ 0, & \text{其他}\end{cases}$$

(3) 当 $0<x<1$ 时

$$f_{Y|X}(y|x)=\frac{f(x,y)}{f_X(x)}=\begin{cases}\dfrac{x+y}{x+\frac{1}{2}}, & 0<y<1\\ 0, & 其他\end{cases}$$

$$f_{X|Y}(x|0.5)=\frac{f(x,0.5)}{f_Y(0.5)}=\begin{cases}x+\frac{1}{2}, & 0<x<1\\ 0, & 其他.\end{cases}$$

2. (10分)已知随机变量X的概率密度函数为

$$f(x)=\begin{cases}x+0.5, & 0<x<1;\\ 0, & 其他.\end{cases}$$

求$E(X),E(X^2),D(X)$.

解: $EX=\int_0^1 x(x+0.5)\,dx=\frac{1}{3}+0.5\cdot\frac{1}{2}=\frac{1}{3}+\frac{1}{4}=\frac{7}{12}$

$E(X^2)=\int_0^1 x^2(x+0.5)\,dx=\frac{1}{4}+0.5\cdot\frac{1}{3}=\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$

$D(X)=E(X^2)-(EX)^2=\frac{5}{12}-\frac{49}{144}=\frac{11}{144}$

3. (30分) 已知随机变量(X,Y)的分布律为

Y \ X	0	1	2
0	0	$\frac{1}{18}$	$\frac{4}{9}$
1	$\frac{1}{9}$	$\frac{7}{18}$	0

(1) 求X的分布律, 以及$E(X),D(X)$;

(2) 求Y的分布律, 以及$E(Y),D(Y)$;

(3) 求$E(XY),cov(X,Y),\rho_{X,Y},D(X+Y)$;

(4) 求$Z_1=X+Y$的分布律, $Z_2=XY$的分布律.

解: (1) $X\sim\begin{pmatrix}0 & 1 & 2\\ \frac{1}{9} & \frac{4}{9} & \frac{4}{9}\end{pmatrix}\Rightarrow X\sim b(2,\frac{2}{3})$

$EX=2\cdot\frac{2}{3}=\frac{4}{3}$, $DX=2\cdot\frac{2}{3}\cdot\frac{1}{3}=\frac{4}{9}$

(2) $Y \sim \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \Rightarrow EX = \frac{1}{2}, \quad DX = \frac{1}{4}$

(3) $E(XY) = \frac{7}{18}$

$$cov(XY) = E(XY) - EX \cdot EY = \frac{7}{18} - \frac{4}{3} \cdot \frac{1}{2} = -\frac{5}{18}$$

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sqrt{DX}\sqrt{DY}} = \frac{-\frac{5}{18}}{\frac{2}{3} \cdot \frac{1}{2}} = -\frac{5}{6}$$

$$D(X+Y) = DX + DY + 2cov(X,Y)$$
$$= \frac{4}{9} + \frac{1}{4} - \frac{5}{9} = \frac{1}{4} - \frac{1}{9} = \frac{5}{36}$$

(4) $Z_1 = X+Y \sim \begin{pmatrix} 1 & 2 \\ \frac{1}{18}+\frac{1}{9} & \frac{4}{9}+\frac{7}{18} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}$

$Z_2 = X \cdot Y \sim \begin{pmatrix} 0 & 1 \\ 1-\frac{7}{18} & \frac{7}{18} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{11}{18} & \frac{7}{18} \end{pmatrix}$

4. (20分) 已知$X \sim N(5,3^2), Y \sim N(0,4^2)$, X,Y独立.

(1) 求$P(X<Y)$;

(2) 已知$(X-Y,X)$服从二维正态分布$N(\mu_1,\mu_2,\sigma_1^2,\sigma_2^2,\rho)$, 求这里的五个参数$\mu_1,\mu_2,\sigma_1^2,\sigma_2^2,\rho$.

解: (1) $X-Y \sim N(5, 5^2)$

$$\Rightarrow P(X<Y) = P(X-Y<0) = P\left(\frac{X-Y-5}{5} < \frac{0-5}{5}\right)$$

$$= \Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

(2) $\mu_1 = E(X-Y) = 5$, $\sigma_1^2 = D(X-Y) = 5^2$

$\mu_2 = EX = 5$, $\sigma_2^2 = DX = 3^2$

$$\operatorname{cov}(X-Y, X) = \operatorname{cov}(X,X) - \operatorname{cov}(Y,X)$$
$$= 3^2 - 0 = 9$$

$$\rho = \frac{\operatorname{cov}(X-Y, X)}{\sigma_1 \sigma_2} = \frac{9}{5 \cdot 3} = \frac{3}{5}$$

5. (10分) 已知$X \sim b(40000, 0.5)$, 试用中心极限定理近似计算$P(X > 20300)$, $P(X > 30000)$, $P(|X - 20000| < 200)$.

解: $EX = 20000$, $DX = 10000$

$X \overset{近似}{\sim} N(20000, 100^2)$

$$P(X > 20300) = P\left(\frac{X-20000}{100} > \frac{20300-20000}{100}\right)$$
$$= 1 - \Phi(3) = 1 - 0.9987 = 0.0013$$

$$P(X > 30000) = P\left(\frac{X-20000}{100} > \frac{30000-20000}{100}\right)$$
$$= 1 - \Phi(100) \approx 1 - 1 = 0$$

$$P(|X-20000| < 200) = P\left(\left|\frac{X-20000}{100}\right| < 2\right)$$
$$= \Phi(2) - \Phi(-2) = \Phi(2) - [1 - \Phi(2)]$$
$$= 2\Phi(2) - 1 = 0.9544$$